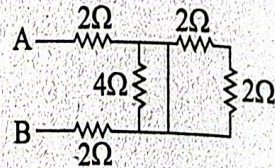
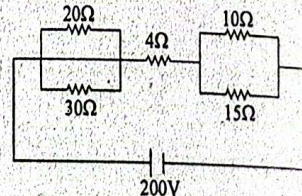
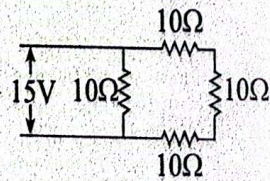


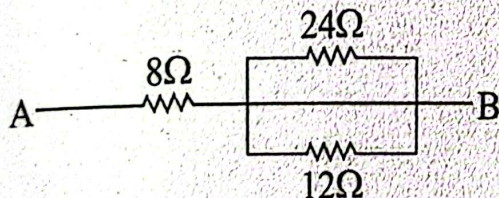
Long Multiple Choice Questions (LMCQs)

- The masses of three wires of copper are in ratio 1:2:3 and their lengths are in ratio 3:2:1. The ratio of their electrical resistances is [IOE 2078]
 - 1:2:3
 - 3:2:1
 - 1:6:27
 - 27:6:1
- The resistance of a conductor at 20°C is 3.15Ω and at 100°C is 3.75Ω . The temperature coefficient of resistance is [IOE 2078]
 - $0.0020^\circ\text{C}^{-1}$
 - $0.0022^\circ\text{C}^{-1}$
 - $0.0025^\circ\text{C}^{-1}$
 - $0.0028^\circ\text{C}^{-1}$
- Two resistors ' $2R$ ' and ' $4R$ ' are connected in parallel and this combination is connected in series with another resistor ' R '. If this whole circuit is connected to a battery of emf ' E ' with negligible internal resistance, the current through ' $4R$ ' resistor is [IOE 2077]
 - $\frac{3E}{7R}$
 - $\frac{7E}{3R}$
 - $\frac{E}{7R}$
 - $\frac{3E}{28R}$
- ABCDEF is a regular hexagon with each side having resistance of 6Ω . The equivalent resistance across any two adjacent corners is [IOE 2077]
 - $\frac{5}{6}\Omega$
 - 5Ω
 - 9Ω
 - 30Ω
- A wire of 16Ω is bent into form of square and two opposite vertices are joined with wire of 16Ω . The equivalent resistance across other two opposite vertices is [IOE 2077]
 - 32Ω
 - 16Ω
 - 8Ω
 - 4Ω
- The length of cylindrical wire is 5 m & its inner and outer radius are 0.5 cm and 1 cm respectively. The resistivity of wire is $3.5 \times 10^{-5}\Omega\text{m}$ then resistance of wire is [IOE 2076]
 - $74.27\mu\Omega$
 - $7.42\text{ m}\Omega$
 - 0.74Ω
 - $0.74\text{ m}\Omega$
- The equivalent resistance across A and B of given circuit is [IOE 2075]
 
 - 2Ω
 - 4Ω
 - 6Ω
 - 8Ω
- The current passing through the circuit is [IOE 2075]
 
 - 5A
 - 10A
 - 15A
 - 20A
- The supply current in the circuit is [IOE 2075]
 
 - 0.5 A
 - 1A
 - 1.5 A
 - 2A
- A wire of length 50 cm and area 1 mm^2 is when applied with a source of 2 V , current of 4 A passes through it. The resistivity of wire is [IOE 2075]
 - $10^{-2}\Omega\text{m}$
 - $10^{-4}\Omega\text{m}$
 - $10^{-6}\Omega\text{cm}$
 - $10^{-8}\Omega\text{m}$

11. In wheatstone bridge $P = 3\Omega$, $Q = 2\Omega$, $R = 6\Omega$ and $S = 8\Omega$. The value of shunt that should be connected to arm 'S' to obtain balance condition is [IOE 2074]

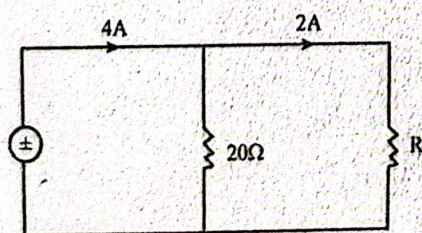
(a) 2Ω (b) 4Ω
(c) 8Ω (d) 16Ω

12. The equivalent resistance between A and B of the given circuit is [IOE 2074]



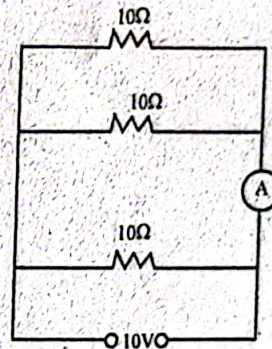
(a) zero (b) 8Ω
(c) 16Ω (d) 3Ω

13. The value of 'R' in the given circuit is [IOE 2073]



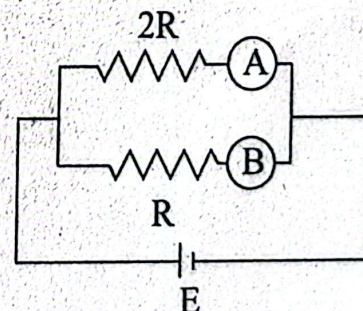
(a) 10Ω (b) 20Ω
(c) 30Ω (d) 40Ω

14. The current through the ammeter in the given circuit is [IOE 2073]



(a) zero (b) 1 A
(c) 2 A (d) 3 A

15. In the given circuit, reading of ammeter 'A' is 5A. The reading of ammeter 'B' is [IOE 2072]



(a) 0 (b) 2.5A
(c) 5A (d) 10A

16. A parallel combination of three resistors take a current of 4A from a 16V source. The two resistors are 10Ω and 15Ω , third resistor will have a value of [IOE 2070]

(a) 12Ω (b) 16Ω
(c) 20Ω (d) 24Ω

Answers

3. The capacity of cell is measured in

[IOE 2077]

- (a) amperes (b) ampere-hours
(c) watts (d) watt-hours

4. Kirchhoff's voltage law is based on

[IOE 2077]

- (a) conservation of charge
(b) conservation of voltage

- (c) conservation of mass
(d) conservation of energy

5. The emf of a cell when it is open circuited is [IOE 2072]

- (a) E (b) $\frac{E}{2}$
(c) $2E$ (d) zero

Long Multiple Choice Questions (LMCQs)

1. Three cells ($2V$, 1Ω), ($5V$, 0.5Ω) and ($3V$, 1Ω) are connected in series and connected to an external resistance of 7.5Ω . The current drawn by external resistance is

[IOE 2078]

- (a) $0.5A$ (b) $1A$
(c) $1.5A$ (d) $2A$

2. A cell of emf $2V$ and internal resistance 0.1Ω is charged by a current of $5A$. The terminal potential difference of cell is

[IOE 2078, 2077]

- (a) $1.5V$ (b) $1.8V$
(c) $2.5V$ (d) $3V$

3. The current through a circuit when two resistors of 6Ω and 12Ω are connected in parallel to a source emf $5V$ and internal resistance of 2Ω will be [IOE 2078]

- (a) $0.53A$ (b) $0.75A$
(d) $0.83A$ (c) $1A$

4. Four cells each $2V$, 2Ω are connected in parallel and connected to an external resistor of 2Ω . The current through external resistor is [IOE 2076]

- (a) $0.2A$ (b) $0.4A$
(c) $0.6A$ (d) $0.8A$

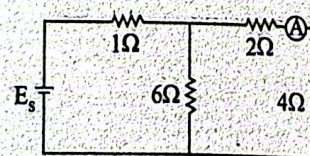
5. Emf of a car battery is $12V$ and its internal resistance is $5 \times 10^{-2}\Omega$. A current of $60A$ flows to starting motor when engine is started. The terminal p.d. of battery is

- (a) $15V$ (b) $12V$
(c) $9V$ (d) $6V$

6. A cell of internal resistance 0.5Ω when connected to a resistor provides $2V$ and $1A$. The e.m.f. of cell is [IOE 2074]

- (a) $3V$ (b) $2.5V$
(c) $2V$ (d) Zero

7. The value of e.m.f. of supply voltage E_s is the given circuit if ammeter reads $2A$ is [IOE 2074]



- (a) $10V$ (b) $12V$
(c) $14V$ (d) $16V$

8. The emf of a cell is $2V$. When a resistance of 10Ω is connected across it, terminal potential difference is found to be $1.5V$. Internal resistance of cell is [IOE 2074]

- (a) 0.3Ω (b) 1.33Ω
(c) 2Ω (d) 3.33Ω

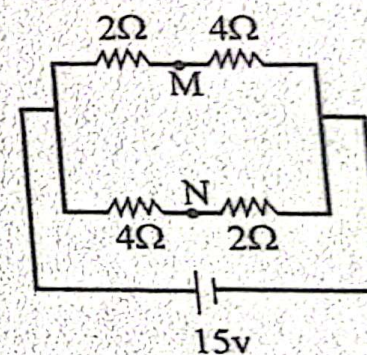
9. The emf of a cell is $10V$ and internal resistance is 1Ω . If $10A$ current is supplied to energize it, its terminal p.d. is [IOE 2073]

- (a) Zero (b) $10V$
(c) $15V$ (d) $20V$

10. An electric cell of e.m.f. 12 V is connected to a $6\ \Omega$ resistor. When resistance is increased by $4\ \Omega$, the circuit current decreases by 0.5 A. The internal resistance of cell is [IOE 2073]

- (a) $1\ \Omega$ (b) $2\ \Omega$
(c) $3\ \Omega$ (d) $4\ \Omega$

11. The potential difference between point M and N of the give circuit is [IOE 2070]



- (a) 5V (b) 7.5V
(c) 10V (d) 12.5V

Answers

8. A voltmeter reads 80 V and has resistance per volt of $100\ \Omega$. The current through voltmeter is [IOE 2076]
- (a) 10 mA (b) 10 A
(c) 1 mA (d) 1 A

9. The current sensitivity of a galvanometer is increased, its voltage sensitivity [IOE 2075]
- (a) Increases (b) Decreases
(c) Remains same (d) None

Long Multiple Choice Questions (LMCQs)

1. A 10m long potentiometer wire with resistance $2\ \Omega/\text{m}$ is connected in series with battery of emf 3V and $10\ \Omega$ resistance. The potential gradient across potentiometer wire is [IOE 2078, 2076]
- (a) 2 V/m (b) 1 V/m
(c) 0.2 V/m (d) 0.1 V/m

2. If 2% of the main current is passed through galvanometer of resistance $49\ \Omega$. The shunt resistance used is [IOE 2076]
- (a) $0.1\ \Omega$ (b) $1\ \Omega$
(c) $0.2\ \Omega$ (d) $2\ \Omega$